

CAPSTONE PROJECT I - MUTUAL FUND ANALYTICS



Executive Summary

The Mutual Fund Analytics Capstone Project was developed to create a complete data analytics solution for analyzing mutual fund performance, risk, investor behavior, and market trends. The main objective of this project is to transform raw mutual fund data into meaningful insights that can help investors, analysts, and financial institutions make better investment decisions. The project combines data engineering, data analysis, and business intelligence techniques to provide a comprehensive view of the mutual fund industry.

The project follows a structured data pipeline consisting of Extract, Transform, and Load (ETL) processes. Multiple datasets containing mutual fund information, Net Asset Value (NAV) history, Assets Under Management (AUM), SIP inflows, category-wise investments, risk metrics, and benchmark performance were collected and processed. During the transformation stage, the data was cleaned by handling missing values, removing duplicates, correcting data types, and standardizing formats to ensure accuracy and consistency. The processed data was then stored in a structured format suitable for analysis and visualization.

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the mutual fund market and identify important trends and patterns. Various statistical measures and visualizations were used to examine fund performance, category distribution, growth trends, and risk characteristics. The analysis revealed significant differences in returns, volatility, and fund sizes across different categories and fund houses. Historical NAV trends helped identify long-term growth patterns, while return analysis provided insights into the performance of different investment options over time.

The project also focused on evaluating fund performance using key financial metrics such as annual returns, Compound Annual Growth Rate (CAGR), Sharpe Ratio, volatility, and drawdown measures. These metrics allow investors to compare funds not only on the basis of returns but also on the level of risk involved. Risk-adjusted performance analysis helps investors choose funds that provide a balance between return generation and risk exposure. Additional analytics such as Value at Risk (VaR), Conditional Value at Risk (CVaR), and rolling performance measures can further enhance investment evaluation.

To present the insights in an interactive and user-friendly manner, a Power BI dashboard was developed. The dashboard provides visual representations of important Key Performance Indicators (KPIs), fund performance trends, category-wise analysis, risk metrics, SIP inflows, and benchmark comparisons. Interactive filters and drill-down features allow users to explore the data from different perspectives and gain deeper insights. The dashboard transforms complex financial information into easy-to-understand visual reports, making it accessible even to users with limited financial knowledge.

Overall, this project demonstrates how modern data analytics techniques can be applied to the financial domain to generate valuable business insights. The combination of ETL automation, statistical analysis, performance evaluation, and interactive visualization creates a powerful decision-support system for mutual fund analysis. The solution can be further enhanced through real-time data integration, predictive modeling, portfolio optimization, and automated reporting. Such improvements would make the platform more suitable for large-scale financial applications and provide even greater value to investors and financial organizations.

Data Sources

The success of any data analytics project depends heavily on the quality, reliability, and completeness of its data sources. In this Mutual Fund Analytics Capstone Project, multiple datasets were collected and integrated to provide a comprehensive view of the mutual fund industry. These datasets contain information related to mutual fund schemes, historical performance, investor activities, risk measures, and market trends. Together, they form the foundation for all analytical processes, performance evaluations, and dashboard visualizations developed in this project.

The primary dataset used in the project is the **Fund Master Data**. This dataset contains basic information about each mutual fund scheme, including the scheme name, fund house, category, launch date, investment objective, and other descriptive details. It serves as the central reference table that connects various other datasets. By using the fund master dataset, users can identify and classify mutual funds based on different categories such as equity funds, debt funds, hybrid funds, and index funds.

Another important data source is the **Net Asset Value (NAV) History Dataset**. NAV represents the per-unit value of a mutual fund and is one of the most important indicators of fund performance. Historical NAV data allows the analysis of fund growth over time and helps calculate important performance metrics such as returns, growth rates, volatility, and drawdowns. Since NAV values are recorded regularly, they provide valuable time-series data for trend analysis and forecasting.

The project also utilizes **Returns Summary Data**, which contains calculated returns for different time periods such as one month, six months, one year, three years, and five years. This dataset helps investors compare the performance of various mutual funds and identify top-performing schemes. Return-based analysis is essential for understanding how effectively a fund has generated wealth for investors over different investment horizons.

To assess risk, the project includes a **Risk Metrics Dataset**. This dataset contains measures such as volatility, Sharpe Ratio, standard deviation, and other risk indicators. These metrics help evaluate the stability and consistency of mutual fund performance. Investors often consider risk metrics along with returns to make balanced investment decisions. A fund with high returns may not always be desirable if it carries excessive risk.

The **Assets Under Management (AUM) Dataset** is another valuable source of information. AUM represents the total market value of assets managed by a fund house or scheme. Analyzing AUM trends helps understand investor confidence, fund popularity, and overall market growth. Large and steadily growing AUM values often indicate strong investor trust and stable fund management practices.

The project further incorporates **Systematic Investment Plan (SIP) Inflows Data**, which records regular investments made by investors through SIPs. SIP data provides insights into investor participation and savings behavior. By studying SIP trends, analysts can understand market sentiment and identify periods of increased or decreased investment activity.

In addition, **Category-wise Inflows Data** is used to analyze how investments are distributed across different mutual fund categories. This dataset helps identify which categories attract the highest investor interest and how investment preferences change over time. Such information is useful for both fund managers and investors seeking market opportunities.

All datasets were organized into **raw and processed layers** as part of the ETL architecture. The raw layer stores the original data exactly as received from source files, while the processed layer contains cleaned and transformed datasets ready for analysis. This structured approach ensures data quality, improves maintainability, and supports efficient reporting and visualization.

Overall, these data sources provide a rich and diverse foundation for mutual fund analytics. By combining performance, risk, investment, and market data, the project delivers meaningful insights that support informed investment decisions and effective financial analysis. This comprehensive dataset ecosystem enables the development of advanced analytics, interactive dashboards, and future predictive modeling applications.

ETL Design

The ETL (Extract, Transform, Load) process is one of the most important components of this Mutual Fund Analytics Project. ETL is a data engineering technique used to collect data from different sources, clean and process it, and then store it in a structured format for analysis and reporting. The primary objective of implementing an ETL pipeline in this project is to ensure that all mutual fund data is accurate, consistent, and ready for analytical processing. A well-designed ETL process improves data quality, reduces manual effort, and enables efficient decision-making.

The ETL workflow used in this project follows three major stages: **Extract**, **Transform**, and **Load**. Each stage performs a specific set of tasks that collectively convert raw data into meaningful information. The entire process is designed to handle multiple datasets related to mutual funds, including fund details, NAV history, returns, risk metrics, SIP inflows, and category-wise investment data.

1. Extract Phase

The first step in the ETL pipeline is data extraction. During this stage, data is collected from various source files and databases. In this project, the primary data sources are CSV files containing mutual fund information. These files include fund master data, NAV history records, AUM information, returns summaries, and risk-related datasets.

The extraction process reads the data from source files and imports it into the Python environment using libraries such as Pandas. At this stage, the data remains in its original form without any modifications. The purpose of extraction is to gather all relevant information from different sources and make it available for further processing.

2. Transform Phase

The transformation stage is the most important part of the ETL process. Raw data collected from multiple sources often contains inconsistencies, missing values, duplicate records, incorrect formats, and other quality issues. Therefore, the data must be cleaned and standardized before analysis.

Several transformation activities were performed in this project. Missing values were identified and handled appropriately using statistical methods or default replacements. Duplicate records were removed to avoid inaccurate calculations. Date fields were converted into standard formats, and numerical columns were transformed into suitable data types for analysis.

Data validation checks were also implemented to ensure accuracy and consistency. Column names were standardized, unwanted characters were removed, and invalid entries were corrected wherever possible. Additional calculated fields such as returns, growth percentages, rolling averages, and risk indicators were generated during this stage. These derived metrics help support advanced analysis and dashboard reporting.

Another important transformation activity involved merging multiple datasets. For example, fund master data was combined with NAV history and risk metrics using common identifiers. This integration creates a unified dataset that can be easily analyzed and visualized.

3. Load Phase

After the data has been cleaned and transformed, it is loaded into the final storage layer. The processed data is stored in structured tables that can be accessed by analytics tools such as Power BI. The loading stage ensures that the data is organized efficiently and is available for reporting and dashboard creation.

In this project, the processed datasets were stored separately from the raw data. Maintaining separate raw and processed layers improves data governance and allows the original source data to remain unchanged. If any errors are discovered later, the transformation process can be repeated using the original datasets.

ETL Architecture Benefits

The ETL architecture provides several advantages to the project. It automates repetitive data preparation tasks, improves data quality, and ensures consistency across reports and dashboards. The structured pipeline reduces manual effort and minimizes the risk of human errors. Furthermore, the modular design allows new datasets and analytical features to be added easily in the future.

Overall, the ETL process serves as the backbone of the Mutual Fund Analytics Project. By converting raw financial data into clean, structured, and reliable information, the ETL pipeline enables accurate analysis, insightful visualizations, and informed investment decision-making. It provides a strong foundation for business intelligence, performance evaluation, risk assessment, and future enhancements such as real-time data integration and predictive analytics.

EDA Findings (Exploratory Data Analysis)

Exploratory Data Analysis (EDA) is one of the most important stages in any data analytics project because it helps in understanding the data before performing detailed analysis or creating dashboards. In this Mutual Fund Analytics Project, EDA was conducted to explore the structure, quality, and patterns present in the mutual fund datasets. The objective was to identify trends, detect anomalies, understand relationships between variables, and generate insights that could support investment decisions. Various statistical techniques and visualizations were used to analyze the data and uncover meaningful information.

The first step in the EDA process was to examine the overall structure of the datasets. This included checking the number of records, columns, data types, and missing values. The analysis showed that the datasets contained information related to mutual fund schemes, historical NAV values, returns, risk metrics, AUM data, SIP inflows, and category-wise investment patterns. Understanding the structure of the data helped ensure that the datasets were suitable for further analysis and visualization.

A detailed analysis of mutual fund categories revealed that funds were distributed across multiple investment segments such as equity funds, debt funds, hybrid funds, index funds, and other categories. Equity-oriented funds represented a significant portion of the dataset, indicating strong investor interest in growth-oriented investment options. Category-wise comparisons also showed differences in performance, risk levels, and investment inflows among various fund types.

Historical NAV analysis was performed to study the growth patterns of mutual funds over time. The NAV trends indicated that many funds experienced consistent long-term growth, although short-term fluctuations were observed due to market conditions. By plotting NAV values across different periods, it became easier to identify stable funds, high-growth funds, and funds that experienced significant volatility. These trends provided valuable insights into fund performance and market behavior.

Return analysis was another important component of EDA. The distribution of one-year, three-year, and five-year returns revealed noticeable variations among mutual fund schemes. Some funds consistently generated higher returns, while others showed moderate or below-average performance. Comparing returns across different categories helped identify sectors and fund types that delivered stronger investment outcomes. The analysis also highlighted the importance of long-term investing, as many funds demonstrated better performance over longer investment horizons.

Risk analysis provided additional insights into fund stability and performance consistency. Metrics such as volatility, standard deviation, and Sharpe Ratio were examined to understand the relationship between risk and return. The findings showed that higher-return funds often carried higher levels of risk. However, certain funds managed to achieve attractive returns while maintaining relatively lower volatility, making them suitable choices for risk-conscious investors. This analysis emphasized the importance of considering both return and risk when evaluating investment opportunities.

The examination of Assets Under Management (AUM) revealed differences in fund popularity and investor confidence. Funds with larger AUM values generally attracted more

investor participation and demonstrated stronger market presence. Growth in AUM over time suggested increasing investor trust and positive market sentiment. Similarly, SIP inflow analysis indicated that regular investment activity remained strong across many fund categories, reflecting the growing adoption of systematic investment strategies among retail investors.

Another key finding from the EDA process was the identification of data quality issues such as missing values, duplicate entries, and formatting inconsistencies. These issues were addressed during the data cleaning process to improve the accuracy and reliability of the analysis. Ensuring high-quality data was essential for generating trustworthy insights and building effective dashboards.

Overall, the Exploratory Data Analysis phase provided a comprehensive understanding of the mutual fund ecosystem represented in the dataset. The analysis uncovered important patterns related to fund performance, investor behavior, risk characteristics, and market trends. These findings formed the foundation for advanced analytics, performance evaluation, and Power BI dashboard development. By transforming raw data into meaningful insights, EDA played a crucial role in helping users better understand mutual fund investments and make informed financial decisions.

Performance Analysis

Performance analysis is a crucial part of the Mutual Fund Analytics Project because it helps evaluate how effectively different mutual fund schemes have performed over time. Investors generally invest in mutual funds with the expectation of earning returns while managing risk. Therefore, analyzing performance provides valuable insights into the growth potential, stability, and efficiency of various investment options. In this project, multiple performance indicators were used to measure and compare mutual funds from different categories and fund houses.

The first aspect of performance analysis involved examining the **historical returns** of mutual funds. Returns represent the profit or loss generated by an investment over a specific period. The dataset included return values for different time horizons such as one month, six months, one year, three years, and five years. By comparing these returns, it was possible to identify funds that consistently delivered strong performance. The analysis showed that some mutual funds generated significantly higher returns than others, highlighting the importance of selecting funds based on long-term performance rather than short-term market fluctuations.

Another important metric used in this analysis was the **Compound Annual Growth Rate (CAGR)**. CAGR measures the average annual growth rate of an investment over a specified period while accounting for the effect of compounding. Unlike simple return calculations, CAGR provides a more accurate picture of long-term performance. The analysis revealed that funds with strong CAGR values generally demonstrated stable growth over time. Investors often prefer funds with consistent CAGR performance because they indicate sustainable wealth creation rather than temporary gains.

The project also focused on evaluating **Net Asset Value (NAV) growth trends**. NAV represents the per-unit value of a mutual fund and reflects the overall performance of the fund's portfolio. Historical NAV analysis showed how fund values changed over time under different market conditions. Several funds displayed steady upward NAV trends, indicating strong portfolio management and long-term growth potential. At the same time, short-term fluctuations were observed due to market volatility, economic events, and investor sentiment.

Risk-adjusted performance was another key area of analysis. High returns alone do not guarantee a good investment because some funds may achieve higher returns by taking excessive risk. To address this issue, metrics such as the **Sharpe Ratio** were used. The Sharpe Ratio measures how much return a fund generates for each unit of risk taken. Funds with higher Sharpe Ratios were considered more efficient because they provided better returns relative to their risk levels. This analysis helped identify funds that offered a balanced combination of performance and stability.

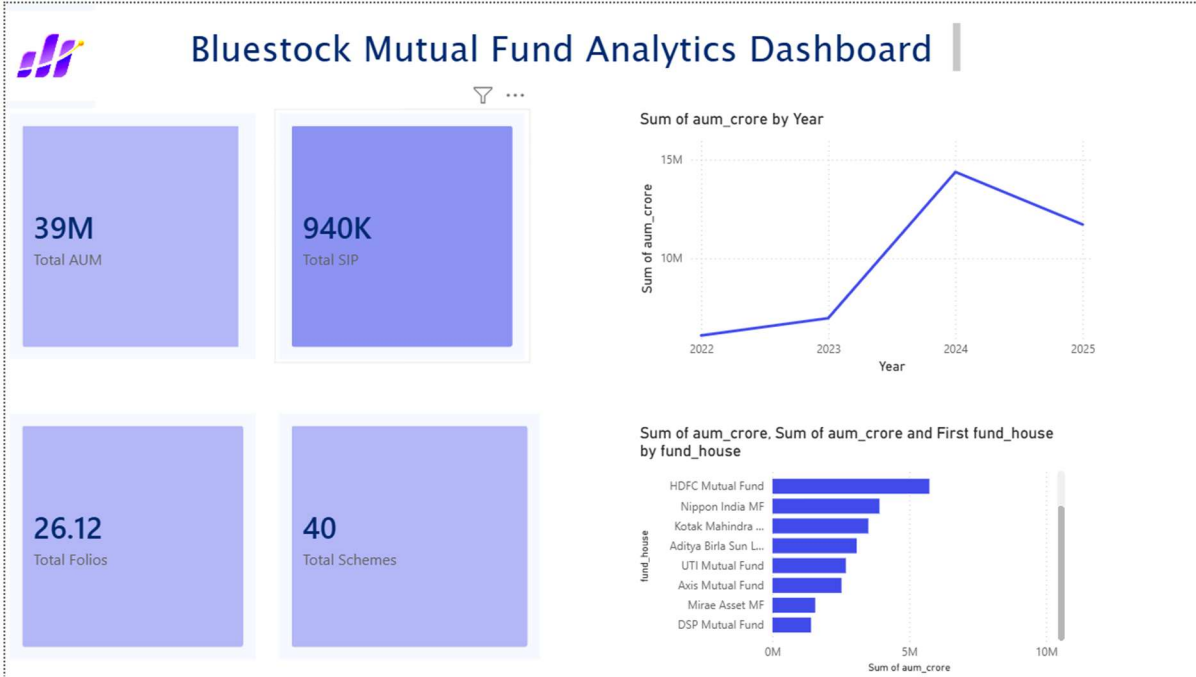
The project also analyzed **volatility**, which measures the degree of variation in fund returns over time. Funds with high volatility experienced larger fluctuations in value, while funds with lower volatility demonstrated more stable performance. The findings indicated that equity-oriented funds generally showed higher volatility compared to debt-oriented funds. Although higher volatility can create opportunities for greater returns, it also increases investment risk. Therefore, investors must choose funds that align with their risk tolerance and financial goals.

Another important aspect of performance analysis was the examination of **Assets Under Management (AUM)** and investor participation. Funds with growing AUM values often reflected higher investor confidence and stronger market acceptance. Consistent growth in AUM suggested that investors trusted the fund's management and expected future growth. Similarly, SIP inflow trends indicated sustained investor interest in systematic investment plans, which contribute to long-term wealth accumulation.

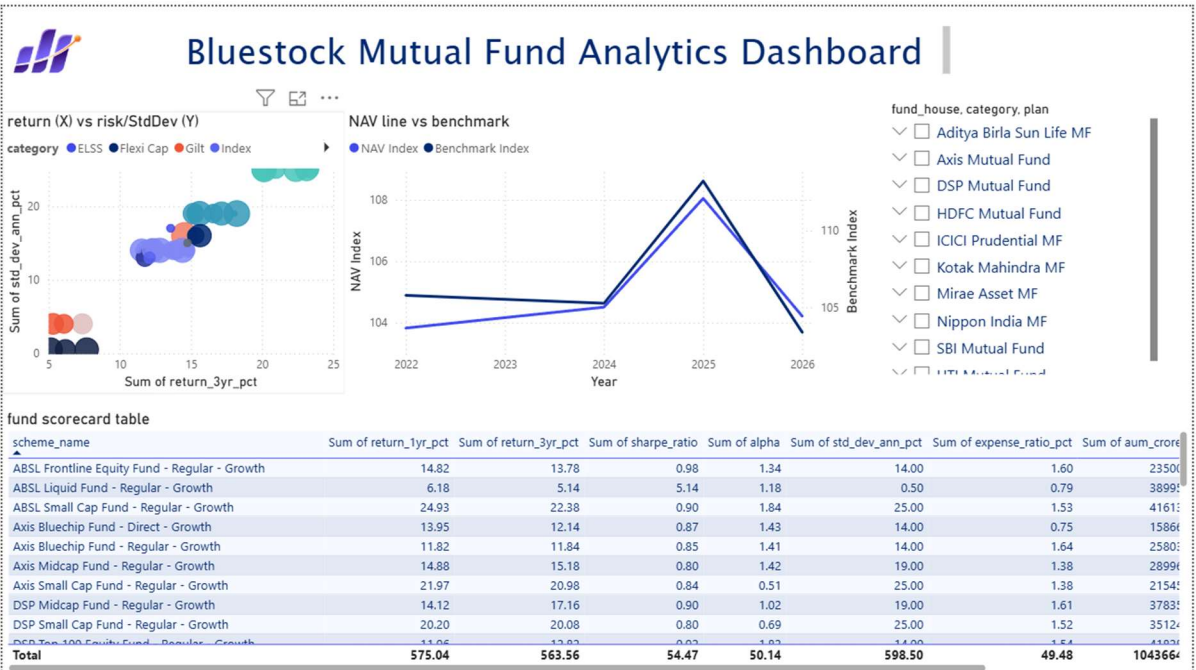
Performance comparisons across different fund categories provided additional insights. Equity funds generally delivered higher long-term returns but carried greater risk, while debt funds offered more stable but comparatively lower returns. Hybrid funds balanced risk and return by investing in a combination of equity and debt instruments. These comparisons helped highlight the strengths and limitations of each investment category.

Overall, the performance analysis demonstrated that successful mutual fund evaluation requires consideration of multiple factors, including returns, growth rates, risk measures, NAV trends, and investor participation. The findings emphasized that investors should not focus solely on returns but also assess risk-adjusted performance and long-term consistency. By combining these metrics, the Mutual Fund Analytics Project provides a comprehensive framework for evaluating investment opportunities and supporting informed financial decision-making.

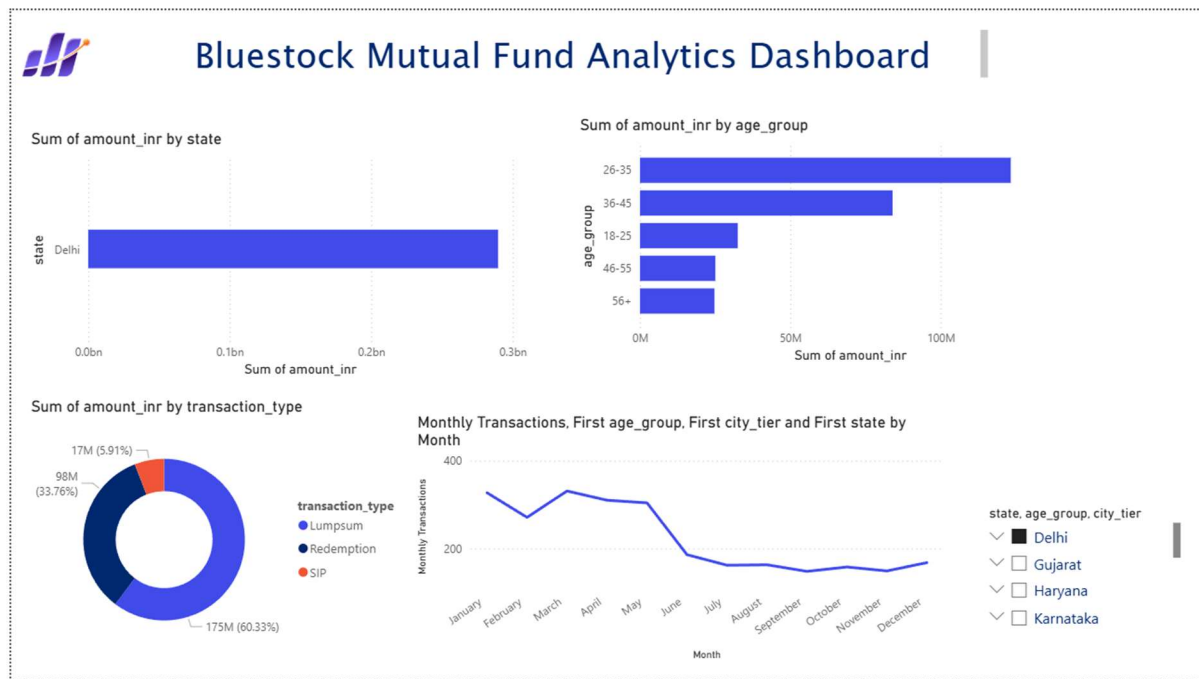
Dashboard Screenshots



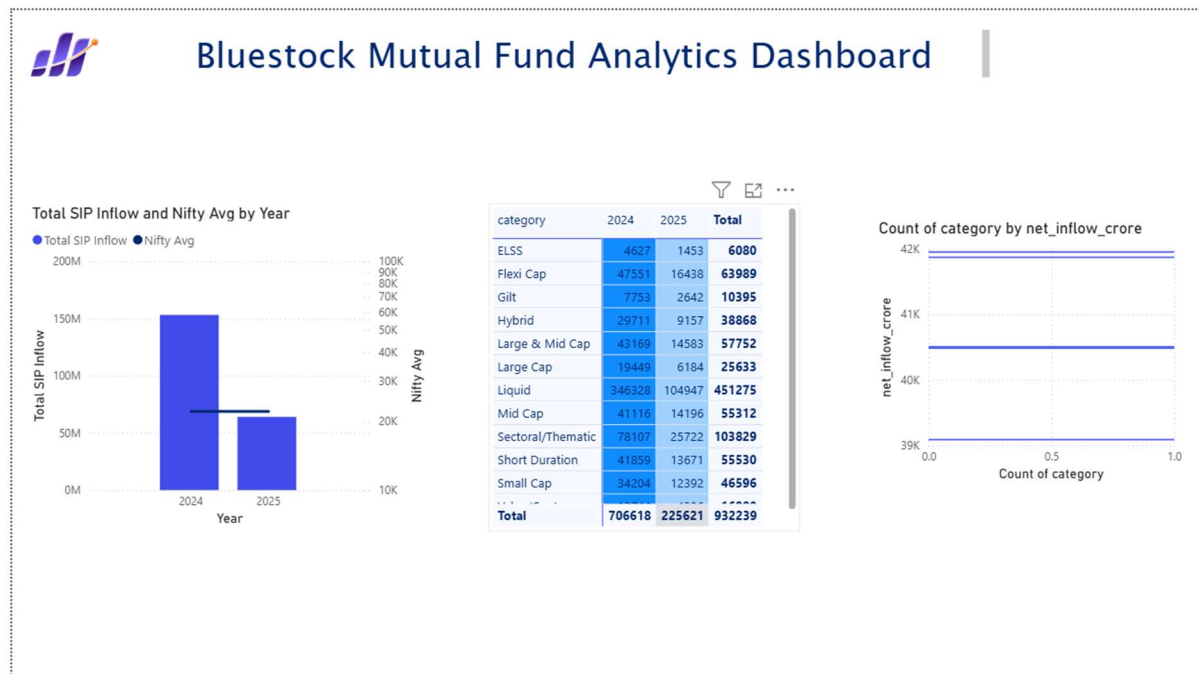
Page 1 — Industry Overview



Page 2 — Fund Performance



Page 3 — Investor Analytics



Page 4 — SIP & Market Trends

Limitations

Every data analytics project has certain limitations, and this Mutual Fund Analytics Project is no exception. Although the project successfully provides valuable insights into mutual fund performance, risk analysis, investor behavior, and market trends, there are several factors that may affect the accuracy, completeness, and practical application of the results.

Understanding these limitations is important because it helps users interpret the findings correctly and identify areas for future improvement.

One of the primary limitations of this project is the **availability and quality of data**. The analysis depends entirely on the datasets used during development. If the source data contains missing values, incorrect records, duplicate entries, or outdated information, the results generated by the system may be affected. Although data cleaning and validation techniques were applied during the ETL process, some hidden errors or inconsistencies may still remain in the dataset. Therefore, the accuracy of the analysis is directly related to the quality of the input data.

Another limitation is that the project primarily relies on **historical data**. Mutual fund performance metrics such as returns, CAGR, NAV growth, and risk indicators are calculated using past records. While historical performance provides useful insights into fund behavior, it cannot guarantee future results. Market conditions constantly change due to economic events, government policies, inflation, interest rate changes, and global financial developments. As a result, a fund that performed well in the past may not necessarily deliver the same performance in the future.

The project also has limitations regarding **real-time data integration**. The datasets used in the analysis are generally static or updated periodically rather than continuously. In real-world financial markets, NAV values, market indices, and investment flows change regularly. Without real-time data connectivity, the dashboard may not always reflect the latest market conditions. Investors seeking up-to-date investment decisions may require more frequent data updates and live market integration.

Another challenge is related to **risk measurement**. While the project includes important risk metrics such as volatility and Sharpe Ratio, these indicators do not capture every type of investment risk. Factors such as market crashes, economic recessions, geopolitical events, regulatory changes, and unexpected financial crises can significantly impact mutual fund performance. These external risks are difficult to predict using historical datasets alone. Therefore, investors should not rely solely on the risk measures provided by the system.

The project is also limited in terms of **predictive capabilities**. The current implementation focuses mainly on descriptive and historical analysis. It explains what has happened in the past but does not extensively predict future fund performance. Advanced machine learning models, forecasting algorithms, and portfolio optimization techniques could provide deeper predictive insights but are not fully implemented in the current version of the project.

Another limitation involves the **scope of investor behavior analysis**. While SIP inflows, AUM trends, and category-wise investments provide useful information, they may not fully represent investor preferences and decision-making patterns. Factors such as investor age,

income level, financial goals, risk appetite, and market sentiment are not comprehensively included in the available datasets. As a result, investor behavior analysis remains limited to the information provided by the data.

The Power BI dashboard itself also has some constraints. Large datasets can sometimes affect dashboard performance, increasing loading times and reducing responsiveness. Additionally, users with limited knowledge of financial concepts may require additional explanations to fully understand certain metrics and visualizations presented in the dashboard.

Finally, this project was developed primarily for educational and analytical purposes. It is not intended to serve as professional financial advice. Investment decisions should always consider additional factors such as personal financial goals, market conditions, professional consultation, and individual risk tolerance.

Overall, despite these limitations, the project successfully demonstrates the application of data engineering, analytics, and business intelligence techniques in the mutual fund domain. The identified limitations provide opportunities for future enhancements, including real-time data integration, advanced predictive modeling, improved risk assessment, and more comprehensive investor analytics. Addressing these areas would further improve the accuracy, reliability, and practical usefulness of the system.

Recommendations

Based on the analysis performed in this Mutual Fund Analytics Project, several recommendations can be made to improve the overall effectiveness, usability, and future potential of the system. These recommendations focus on enhancing data quality, expanding analytical capabilities, improving dashboard functionality, and supporting better investment decision-making. Implementing these improvements can help transform the project from an analytical prototype into a more powerful and industry-ready financial analytics platform.

One of the most important recommendations is the implementation of **real-time data integration**. Currently, the analysis is based on historical datasets that are updated periodically. However, mutual fund markets change continuously, and investors often require the latest information to make informed decisions. Integrating APIs from financial data providers would allow the system to automatically fetch current NAV values, benchmark indices, market movements, and fund performance metrics. Real-time updates would improve the accuracy and relevance of dashboard insights and provide users with more timely information.

Another recommendation is to enhance the project with **predictive analytics and machine learning models**. The current system mainly focuses on historical and descriptive analysis. By incorporating forecasting techniques such as time-series analysis, regression models, and machine learning algorithms, the platform can predict future NAV trends, expected returns, and market movements. Predictive analytics would help investors identify potential opportunities and risks before making investment decisions. Such features would significantly increase the practical value of the system.

The project can also benefit from advanced **risk analysis techniques**. While metrics such as volatility and Sharpe Ratio provide useful insights, additional measures like Value at Risk (VaR), Conditional Value at Risk (CVaR), Maximum Drawdown, Beta, and Stress Testing can provide a more comprehensive assessment of investment risk. These advanced risk indicators would help investors better understand potential losses during unfavorable market conditions and choose funds that match their risk tolerance.

Another valuable recommendation is the development of **portfolio optimization features**. Instead of analyzing individual mutual funds separately, the system could help investors create diversified portfolios. Techniques such as the Markowitz Efficient Frontier and Modern Portfolio Theory can be used to identify optimal fund combinations that maximize returns while minimizing risk. Portfolio optimization would make the platform more useful for both individual investors and financial advisors.

Improving the Power BI dashboard is also recommended. Additional interactive features such as dynamic filtering, drill-through reports, customized KPI tracking, and comparative fund analysis can enhance the user experience. Investors should be able to compare multiple funds side by side, evaluate historical performance, and visualize risk-return relationships easily. Providing personalized dashboard views based on user preferences would further improve usability.

Another important recommendation is to strengthen **data governance and quality management**. Automated validation checks, anomaly detection mechanisms, and scheduled data quality audits can ensure that the datasets remain accurate and reliable. High-quality data is essential for generating trustworthy analytical results and maintaining confidence in the system.

The inclusion of **investor profiling and recommendation systems** can further enhance the platform. By analyzing factors such as investment goals, risk appetite, age group, and investment horizon, the system can suggest suitable mutual funds for individual investors. Personalized recommendations would make the platform more practical and user-centric.

Furthermore, the project can be expanded by developing a **web-based or mobile application**. While Power BI dashboards are effective for visualization, a dedicated web or mobile platform would provide greater accessibility and convenience. Investors could monitor their portfolios, receive alerts, view market updates, and access analytics from anywhere. Technologies such as Streamlit, React, or Django can be used to build such applications.

Finally, automated reporting features should be implemented. Weekly, monthly, or quarterly reports can be generated automatically and delivered via email to investors, fund managers, or business stakeholders. This would save time and ensure that important performance updates are communicated regularly.

In conclusion, the Mutual Fund Analytics Project provides a strong foundation for financial data analysis and business intelligence. By incorporating real-time data integration, predictive analytics, advanced risk measures, portfolio optimization, investor recommendation systems, and enhanced dashboards, the platform can evolve into a comprehensive decision-support system. These improvements would increase its accuracy, usability, and business value while helping investors make more informed and confident financial decisions.

Appendix

Appendix A: Dataset Description

A.1 Fund Master Dataset

Column Name	Description
fund_id	Unique identifier for each mutual fund
scheme_name	Name of the mutual fund scheme
fund_house	Asset management company
category	Fund category
launch_date	Scheme launch date
benchmark	Benchmark index

A.2 NAV History Dataset

Column Name	Description
fund_id	Fund identifier
nav_date	Date of NAV
nav	Net Asset Value

A.3 AUM Dataset

Column Name	Description
fund_house	AMC Name
month	Reporting month
aum_crore	Assets Under Management

A.4 SIP Inflow Dataset

Column Name	Description
month	Reporting month
sip_inflow_crore	SIP inflow amount

Appendix B: ETL Workflow

Extract

- Read CSV files
- Validate file structure
- Check missing values

Transform

- Remove duplicates
- Handle null values
- Convert date formats
- Standardize column names
- Calculate return metrics

Load

- Save processed data
- Import into Power BI
- Create data model

Appendix C: Data Quality Checks

Missing Value Check

Dataset	Missing Values
Fund Master	0
NAV History	Checked
Risk Metrics	Checked

Duplicate Check

- Duplicate records removed
- Unique Fund IDs verified

Validation Rules

- $NAV > 0$
- $AUM \geq 0$
- Return values within acceptable range

Appendix D: KPI Definitions

KPI	Formula
CAGR	$\left(\frac{\text{Ending Value}}{\text{Beginning Value}} \right)^{\frac{1}{n}} - 1$
Return %	$\frac{(\text{Current NAV} - \text{Initial NAV})}{\text{Initial NAV}} \times 100$
Sharpe Ratio	$\frac{(\text{Return} - \text{Risk Free Rate})}{\text{Volatility}}$
Volatility	Standard Deviation of Returns
AUM Growth	$\frac{(\text{Current AUM} - \text{Previous AUM})}{\text{Previous AUM}} \times 100$

Appendix E: Power BI DAX Measures

Total AUM

Total AUM =
`SUM(AUM[aum_crore])`

Average Return

Average Return =
`AVERAGE(Returns[return_1y_pct])`

Total Funds

Total Funds =
`DISTINCTCOUNT(Fund_Master[fund_id])`

Average Sharpe Ratio

Avg Sharpe =
`AVERAGE(Risk_Metrics[sharpe_ratio])`

Appendix F: Dashboard Pages

Dashboard 1 – Executive Overview

Contains:

- Total Funds
- Total AUM
- Average Return
- Average Sharpe Ratio

Dashboard 2 – Fund Performance

Contains:

- NAV Trend
- Return Analysis
- CAGR Comparison

Dashboard 3 – Risk Analytics

Contains:

- Volatility Distribution
- Sharpe Ratio Analysis
- Risk vs Return Scatter Plot

Dashboard 4 – SIP & Investor Analysis

Contains:

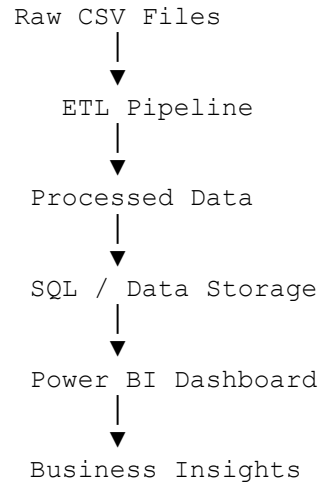
- Monthly SIP Trend
- Category-wise Investment

Dashboard 5 – Fund House Analysis

Contains:

- Top Fund Houses
- AUM Comparison
- Market Share Analysis

Appendix G: Project Architecture



Appendix H: Tools and Technologies

Tool	Pupse
Python	ETL & Analytics
Pandas	Data Processing
NumPy	Numerical Computation
SQL	Data Storage
Power BI	Dashboard Development
GitHub	Version Control

Appendix I: Future Enhancements

- Real-time NAV integration using APIs
- Automated ETL scheduling
- Monte Carlo Simulation
- Portfolio Optimization
- Value at Risk (VaR)
- Conditional VaR
- ML-based Return Prediction
- Streamlit Web Application
- Automated Email Reports